

ALEC SANCHEZ

Security Engineer | Security Architect | Zero Trust | Detection Engineering
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PROFESSIONAL SUMMARY

Security engineer and architect with proven experience designing Zero Trust network environments, operating enterprise SOC toolstacks, and conducting structured adversary simulation. Author of the **OSIM Framework** — a formally governed Zero Trust methodology with 17 machine-verifiable security invariants, ATT&CK-mapped; threat boundaries, and governance-as-code. Diagnosed and patched **three production defects** in OPNsense 26.1.5 core through packet capture analysis and PHP source review, restoring reliable DNS forwarding across segmented VLAN environments. Deployed and integrated Wazuh SIEM, Suricata IDS/IPS, MISP threat intelligence, and Caldera adversary emulation across a five-zone enterprise simulation environment. CompTIA Security+ and PJPT in progress.

CORE SKILLS

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|--------------------------------|--------------------------------|---------------------------------|--------------------------------------|
| • Zero Trust Architecture | • Network Segmentation & VLANs | • Firewall Policy & OPNsense/pf | • IDS/IPS (Suricata) |
| • SIEM & Detection Engineering | • Threat Intelligence (MISP) | • Adversary Emulation (Caldera) | • Incident Response & Runbooks |
| • ATT&CK; Threat Modeling | • Vulnerability Management | • Linux / FreeBSD / Windows | • Terraform · Ansible · Docker · Git |

PROFESSIONAL EXPERIENCE

Security Engineer & Architect — OSIM Framework

Jan 2025 – Present

Independent Security R&D; · github.com/osim-framework

- Designed and implemented OSIM Framework — Zero Trust security architecture methodology with 8 principles, 17 machine-verifiable invariants, ATT&CK-mapped; threat model, and governance-as-code; documented in 32-page professional portfolio
- Architected five-zone VLAN-segmented enterprise environment on OPNsense enforcing default-deny policy with explicit allow rules, alias-based rule construction, DNS confinement, and SHA-256 config drift detection
- Diagnosed and patched 3 production defects in OPNsense 26.1.5 core PHP (unbound.inc) through WAN packet capture, Unbound infrastructure cache analysis, and source review; submitted responsible disclosure with unified diffs to OPNsense project
- Deployed Wazuh SIEM across all five network zones with multi-zone agent telemetry, firewall deny-event correlation, Suricata eve.json ingestion, and 30-day anomaly baselines
- Operated MISP threat intelligence platform with 6 live feeds ingesting 59+ structured events; executed RT-01–RT-03 adversary simulation scenarios via Caldera mapped to ATT&CK; T1611, T1048.003, T1059 with mandatory Wazuh alert evidence
- Identified and disclosed Bug TH-MISP-001 — HTTP Authorization header mismatch breaking TheHive-MISP integration in all default deployments; engineered fix and submitted full disclosure to StrangeBee
- Provisioned Windows Server 2022 AD domain with Group Policy, DNS, DHCP, Sysmon telemetry, and Wazuh host agents; managed infrastructure via Terraform and Ansible on Proxmox VE with tiered NVMe storage

Restaurant Server

Oct 2017 – Feb

2022

Killen's Barbecue, Houston TX

- Developed composure under pressure, real-time prioritization, and cross-team coordination in high-volume operations — directly transferable to SOC environments

SELECTED SECURITY PROJECTS

OSIM Framework — Zero Trust methodology with machine-verifiable invariants, ATT&CK; threat mapping, 4-layer detection model, and governance tooling. Implemented on production hardware.

alecsanchezsecurityportfolio.netlify.app

OPNsense Bug Fixes — Discovered and patched 3 production defects in OPNsense 26.1.5 Unbound DNS configuration generator via packet analysis and PHP source review. github.com/osim-framework/opnsense-unbound-fixes

TheHive-MISP Bug (TH-MISP-001) — Discovered HTTP header mismatch breaking default TheHive-MISP integration. Engineered Apache mod_headers fix and submitted full disclosure to StrangeBee.

EDUCATION

CS Coursework — University of Houston · 2023–25

A.S. Computer Science — San Jacinto College · 2021–23

A.A.S. Business Administration — San Jacinto College · 2015–17

CERTIFICATIONS

- CompTIA Security+ (*In Progress*)
- PJPT — Practical Junior Penetration Tester, TCM Security (*In Progress*)